

AAYAN HAIDER

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PROFILE

Computer science and mathematics graduate from BRAC University with strong experience in computational and statistical research, including machine learning (ML) pipelines, bioinformatics, explainability, and causal inference. Completed and defended an undergraduate thesis on causal artificial intelligence (AI) and epigenetics with ongoing efforts for publication. Interested in teaching and engaging in research on causality, explainability, statistical learning, graphical models, speech processing, and natural language processing.

EDUCATION

B.Sc. in Computer Science; Double major in Mathematics

BRAC University

September 2022 – June 20026

Dhaka, Bangladesh

- **Thesis title:** From DMRs to Disease Diagnosis: A Causal and Inferential Framework for Schizophrenia Using DNA Methylation
- **CGPA:** 3.91 / 4.00
- **Relevant coursework:** Algorithms, Data Structures, Artificial Intelligence, Numerical Methods, Computer Graphics, Database Systems, Statistics & Probability, Linear Algebra, Real Analysis, Graph Theory, Dynamical Systems, Numerical Analysis

Cambridge International AS & A Level

Scholastica

August 2020 – August 2022

Dhaka, Bangladesh

- **GPA:** 4.00 / 4.00

Cambridge International O Level

Scholastica

August 2018 – August 2022

Dhaka, Bangladesh

- **GPA:** 4.00 / 4.00

RESEARCH EXPERIENCE

Undergraduate Researcher

Department of Computer Science and Engineering, BRAC University

June 2025 – June 2026

- **Title:** From DMRs to Disease Diagnosis: A Causal and Inferential Framework for Schizophrenia Using DNA Methylation
- Integrated four GEO DNA methylation datasets comprising 2,728 samples for schizophrenia classification.
- Developed a hierarchical computational framework linking differentially methylated regions (DMRs), genes, and co-expression modules to perform causal discovery across multiple abstraction levels and derive a minimal, causally interpretable feature set.
- Implemented data preprocessing, batch effect correction, feature selection, feature aggregation, and statistical/ML analyses using Python and R.
- Achieved minimal degradation in predictive performance relative to statistically- and explainable-AI-derived feature sets, while enabling biological contextualisation of selected features through causal discovery and domain knowledge.
- Manuscript based on this research titled "Hierarchical Recursive Causal Feature Selection for Explainable Schizophrenia Prediction from DNA Methylation" submitted to *Array*; currently under review.

PROJECTS

Interactive Aquatic Ecosystem Simulation

- Team coursework project
- Developed an animated fish simulation in Python using OpenGL, modelling emergent flocking behaviour through alignment, cohesion, and separation forces.
- Implemented state-dependent predator/prey, food-seeking, obstacle-avoidance, and coordinated gathering behaviours.

TECHNICAL SKILLS

- **Programming:** Python, C/C++
- **Data & Scientific Computing:** NumPy, pandas, SciPy, SymPy, Matplotlib
- **Machine Learning:** scikit-learn, SHAP
- **Statistical & Computational Methods:** Statistical modelling, causal discovery (DirectLiNGAM), causal inference (DoWhy), network analysis (NetworkX)
- **Development & Tools:** Git, GitHub, LaTeX
- **Graphics:** OpenGL

HONOURS & ACHIEVEMENTS

- **VC's List (Latest 2026):** Awarded by BRAC University for earning a semester GPA of 3.91 or above in a 4.00 scale with at least 12 credits taken without any failed or withdrawn courses. Recognised on the VC's List in six semesters between 2022 and 2026.
- **Dean's List (Latest 2025):** Awarded by BRAC University for earning a semester GPA of 3.70 to 3.90, inclusive, on a 4.00 scale with at least 12 credits taken without any failed or withdrawn courses. Recognised on the Dean's List in three semesters between 2024 and 2025.
- **Daily Star Award (2022):** Awarded for excellence in the Cambridge International A Level examinations.
- **Best prefect award (2017):** Awarded for being the best male prefect in grade 7.
- **High achiever awards (2013 – 2019):** Awarded for excellence in school coursework every academic year from grade 4 to grade 9, inclusive.

SCHOLARSHIPS

Merit Scholarship based on BRAC University Academic Results

BRAC University

📅 January 2024 – September 2025

50% scholarship on tuition fees awarded to students with a CGPA between 3.90 and 3.94 (inclusive) on a 4.00 scale.

Scholarship Based on Previous Academic Results

BRAC University

📅 September 2022 – December 2023

25% scholarship on tuition fees awarded to students based on their O level and A level examination results.

LANGUAGES

- Bengali (Native proficiency)
- English (Bilingual proficiency; CEFR C2)